

## Project Design Phase-II

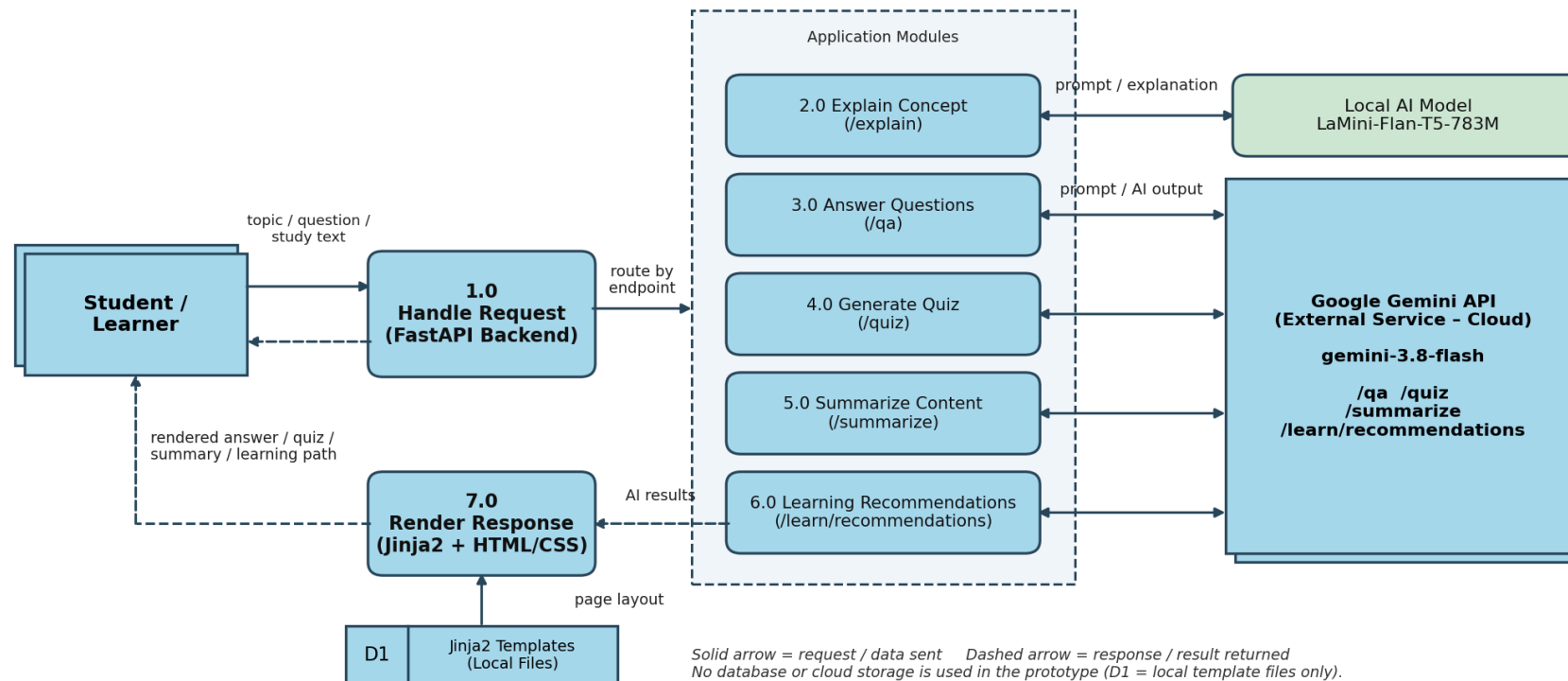
### Data Flow Diagram & User Stories

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 1 October 2026                                     |
| Team ID       | 6ab7f80ad083b0e2ee72b17e                           |
| Project Name  | EduGenie: Google Gemini Powered Learning Assistant |
| Maximum Marks | 4 Marks                                            |

#### Data Flow Diagrams:

Problem (PS-3): A student wants to revise what they have studied and check their understanding, but going through long study materials again and creating practice questions takes a lot of time. EduGenie generates concise revision points and personalized practice questions.

#### DFD Level 1 - EduGenie: Google Gemini Powered Learning Assistant



1. The student opens the EduGenie web page (HTML/CSS) and enters a topic, question or study text.
2. The request goes to the FastAPI backend (Uvicorn), which routes it to the matching feature module.
3. /explain: the local LaMini-Flan-T5-783M model generates a simplified explanation.
4. /qa, /quiz, /summarize, /learn/recommendations: the prompt is sent to the Google Gemini API (gemini-3.8-flash).
5. The AI output is returned to FastAPI, rendered through Jinja2 templates and shown to the student. No data is stored.

## User Stories

User stories for the EduGenie product.

| User Type                 | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                           | Acceptance criteria                                                         | Priority | Release  |
|---------------------------|-------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------|----------|
| Student (Learner)         | Concept Explanation           | USN-1             | As a student, I can enter a topic or concept and get a simple explanation of it.                            | I receive an easy-to-understand explanation on the page.                    | High     | Sprint-1 |
|                           |                               | USN-2             | As a student, I can get explanations generated by the local model so they work without an external service. | /explain returns a result using LaMini-Flan-T5-783M.                        | Medium   | Sprint-1 |
|                           | Question Answering            | USN-3             | As a student, I can ask a question about what I am studying and get a clear answer.                         | I receive a relevant answer from Gemini on the same page.                   | High     | Sprint-1 |
|                           | Summarization                 | USN-4             | As a student, I can paste my study material and get concise revision points.                                | I receive a short summary with the key points.                              | High     | Sprint-2 |
|                           |                               | USN-5             | As a student, I can save time by not re-reading long notes before exams.                                    | Summary is much shorter than the original text and covers main ideas.       | Medium   | Sprint-2 |
|                           | Quiz Generation               | USN-6             | As a student, I can generate practice questions from a topic or my study material.                          | I receive a set of practice questions without writing them myself.          | High     | Sprint-2 |
|                           |                               | USN-7             | As a student, I can check my answers to the quiz to know if I understand the topic.                         | I can see correct answers and judge my understanding.                       | Medium   | Sprint-2 |
|                           | Learning Recommendations      | USN-8             | As a student, I can get suggestions on what to learn next.                                                  | I receive a learning path with next topics to study.                        | Medium   | Sprint-3 |
|                           | Web Interface                 | USN-9             | As a student, I can use all features from one simple web page.                                              | Explain, Q&A, Quiz, Summary and Recommendations are accessible from the UI. | High     | Sprint-1 |
|                           |                               | USN-10            | As a student, I see a clear message if the Gemini service is unavailable.                                   | A friendly error message is shown instead of a crash.                       | Low      | Sprint-3 |
| Administrator / Developer | Setup & Security              | USN-11            | As an administrator, I can keep the Gemini API key in an environment variable.                              | The API key is not exposed in code or in the browser.                       | High     | Sprint-1 |
|                           |                               | USN-12            | As an administrator, I can run the app locally on a Uvicorn server.                                         | The app starts locally and all pages load.                                  | Medium   | Sprint-1 |